

CaptionAI: AI-Powered Social Media Caption Generator

Project Description:

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The Smart Lender – Loan Approval Prediction System is a Machine Learning-based web application designed to help banks and financial institutions make faster and more accurate loan approval decisions. The system analyzes applicant information such as Gender, Married Status, Dependents, Education, Self Employment, Applicant Income, Co-applicant Income, Loan Amount, Loan Amount Term, Credit History, and Property Area to predict whether a loan should be Approved or Rejected. The application uses a trained XGBoost Machine Learning model developed with Scikit-learn and deployed using the Flask framework. It provides real-time predictions through a simple web interface, reducing manual effort, improving decision-making, and minimizing the risk of loan defaults.

Scenarios

Scenario 1: Loan Approval

A customer with a good credit history, stable income, and eligible loan details submits an application. The system predicts Loan Approved.

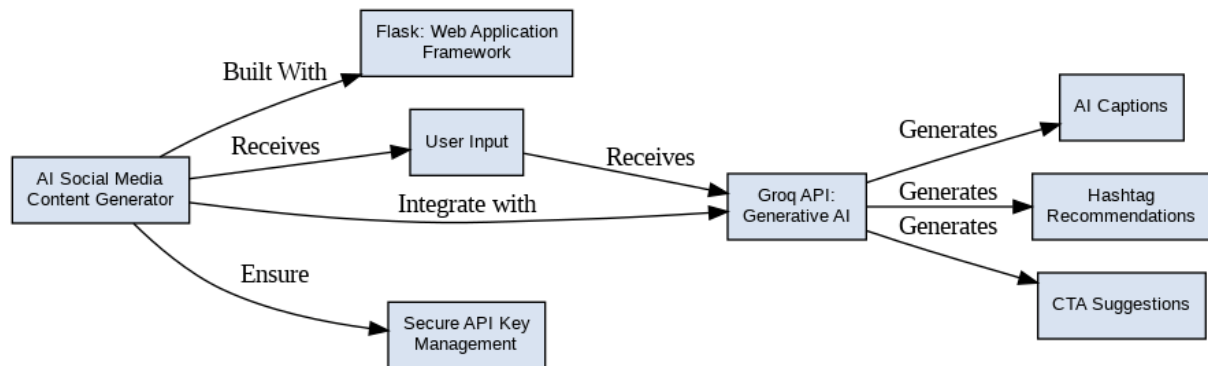
Scenario 2: Loan Rejection

A customer with poor credit history or insufficient income applies for a loan. The system predicts Loan Rejected based on the applicant's details.

Scenario 3: Bank Decision Support

The bank officer enters applicant information into the Smart Lender system. The machine learning model instantly predicts the loan status, helping the officer make a quick and reliable lending decision.

Technical Architecture:



Description: CaptionAI utilizes a modular three-tier architecture to deliver rapid, AI-driven content generation. The frontend provides a responsive user interface built with HTML and CSS while the Flask-based backend orchestrates data validation and prompt construction. It interfaces with the Groq API using the LLaMA 3.3-70B model to generate tailored social media content, with secure deployment managed locally via Flask and exposed publicly through Ngrok.

(Note: If you are using database in your project definitely you need to add ER Diagram along with description)

Pre-requisites:

- **Flask Framework Knowledge:** [Flask Documentation](#)
- **Groq API Familiarity:** [Groq API Documentation](#)
- **HTML, CSS, and JavaScript Skills:** [W3Schools HTML/CSS/JavaScript Tutorials](#)
- **Python Programming Proficiency:** [Python Documentation](#)
- **Version Control with Git:** [Git Documentation](#)
- **Development Environment Setup:** [Flask Installation Guide](#)
- **Ngrok for Public Deployment:** [Ngrok Documentation](#)

Project Workflow:

Milestone 1: Model Selection and Architecture

- **Activity 1.1:** Generate a Groq API key and configure it securely in the project environment.